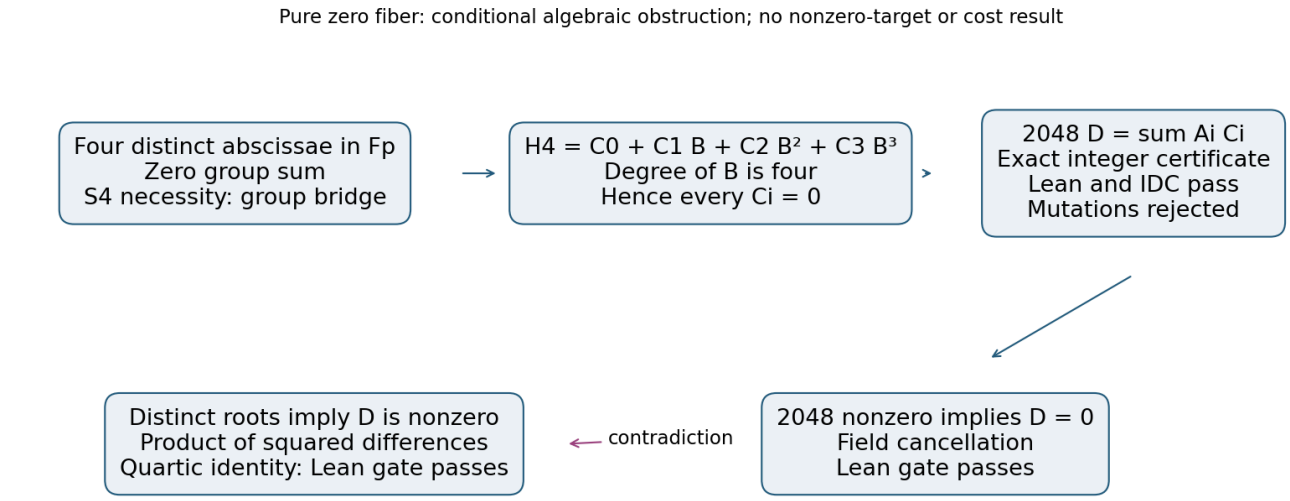
Exact component certificate for the pure zero fiber

The 34-equation, 31-unknown rational search in pure-zero-certificate-v1 finds an exact identity. Independent integer multiplication gives zero remainder and rejects a constant +1 mutation. The actual coefficient certificate (../../results/pure-zero-obstruction/certificate.json) has SHA-256 3d0cfc9cd47e1b0633a0cc4e0fca05256ad6a55e574e7be3831994052479f73f. The separate native replay now passes: exact integer remainder zero, IDC1h09d838b7dca9895f accepted with 32 extra points, and both constant and multiplier mutations rejected. Its unchanged receipt is runs/pure-zero-verify-v1/verified/summary.json. The Lean gate now passes both the component identity and its field cancellation consequence, with standard foundations only and no trusted native declaration.



Statement and readable argument

Let F have characteristic other than 2 or 3, let K contain F, and let B have degree four over F. On the nonsingular curve E: y^2=x^3+B, four finite points whose abscissae are pairwise distinct elements of F cannot have a signed sum equal to O. Signs do not change abscissae.

Write s1,s2,s3,s4 for their elementary symmetric coordinates. The already integer-verified, fully symmetric S4 polynomial is

H4 = C0 + C1*B + C2*B^2 + C3*B^3.

A signed zero relation makes H4 vanish: the finite point obtained by adding the first two signed points supplies a common root of the two S3 quadratics defining the resultant. Pairwise distinct abscissae exclude an infinite intermediate sum. This is the standard summation-polynomial necessity argument; its actual curve/group formalization is not provided by the new coefficient proof. The finite-field definition and group-sum property are stated in Gorla–Massierer, Definition 2.4 and Theorem 2.5 (https://arxiv.org/html/1405.1059), citing Semaev's original construction (https://eprint.iacr.org/2004/031). The new ideal-membership identity and the degree-four coefficient argument are the present derivation; no novelty claim is made for the result.

Since s1,...,s4 lie in F and 1,B,B^2,B^3 are independent over F, all four Ci vanish. With a=s1,b=s2,c=s3,d=s4, the exact search certificate is

2048*D = A0*C0 + A1*C1 + A2*C2 + A3*C3, A0 = -8192, A1 = -64*c + 96*a*b - 24*a^3, A2 = 112*c^2 - 512*b*d + 8*a*b*c + 48*a^2*d, A3 = 137*c^3 - 588*b*c*d + 2*b^3*c + 792*a*d^2 + 3*a*b*c^2.

Here D is the standard quartic discriminant, not an arbitrary target polynomial. Thus D=0 in odd characteristic. But its exact factorization is the product of the six squared pair differences, which is nonzero in the field F when all abscissae are distinct. This is the contradiction. The independent quartic factorization Lean gate (../run-20261024T0252Z-quartic-factored/REPORT.md) already passes; the new component and field-cancellation statements also pass their separate two-theorem gate. No full Lean curve-level theorem is claimed.

For the actual models B=b0*t^3, t^4=d0 irreducible and b0*d0 nonzero, t=B^3/(b0^3*d0^2), so B generates the same degree-four extension. This supplies the stated field-basis hypothesis for those models. Checking the finite model data and embedding, and formalizing this implication, remain distinct from verifying the integer certificate.

Consequences and exclusions

Under the stated hypotheses, zero-fiber four-distinct-factor precomputation in the pure Fp-abscissa base cannot supply relations. Mixed images, nonzero residuals and higher arities remain open. In particular, the 72 doubled zero relations in the first diagnostic do not contradict it; their separate control checks that their folded rows vanish.

The prediction for every pure-L0 zero set in the running 21-case census is an empty signed set. This cross-check is recorded before inspecting the remaining census contents. Any mismatch requires investigation of the exact public model, field-basis, S4 or signed-set implementation; it must not be silently discarded. Repeated factors, infinite factors, other base components, three-factor CM identities, five or more factors and target descent are outside the statement.

The search is native Rust under one thread, the busy reservation, 8 GiB and a 60 s watchdog, with separate compilation. It makes no runtime, asymptotic or novelty claim. Prior S4/symmetry methods remain the baseline.